GaddamGaganMohanReddy

AI Engineer

+1 628-666-4325 | mohan750r@gmail.com | Open to Relocate

PROFESSIONAL SUMMARY

- Senior AI/ML Platform Engineer with 7.5+ years of experience building production-grade LLM agents, agentic workflows, and real- time inference platforms across financial services and healthcare, serving 1M+ users at enterprise scale.
- Deep expertise in multi-agent system architecture - separating retrieval, reasoning, and execution layers for independent scaling - with hands-on, end-to-end ownership from document ingestion through customer-facing product surfaces.
- Proven track record in model fine-tuning using LoRA and RLHF, building continuous evaluation pipelines, and designing feedback loops with domain experts that measurably improve model quality in regulated production environments.
- End-to-end AWS cloud practitioner (primary) with strong Azure depth - complete IaC ownership via AWS CDK, and production deployments on Azure AI Foundry, Databricks, and AKS with zero outages during rollouts.
- Expert in RAG pipeline engineering - advanced chunking strategies, hybrid retrieval, embedding benchmarking, and vector DB optimization - delivering high-precision document retrieval across healthcare policy and financial compliance corpora.
- Strong MLOps and systems reliability discipline - circuit-breaker patterns, graceful degradation, SLI/SLO-driven observability, fallback retrieval strategies, and resilience design maintaining 99.8%+ uptime under upstream LLM failures.
- Compliance-first engineer with zero violations across PCI-DSS and HIPAA regulated systems - enforcing multi-tenant data isolation, PHI leak prevention with 99.7% block rate, and bias detection across all LLM outputs.
- Experienced building intelligent LLM routing and cost optimization layers - selecting between model tiers by query complexity, achieving $42K+ annual inference savings without degrading accuracy or user experience.
- Skilled in designing and running A/B evaluation frameworks for prompt and agent behavior variations at scale, improving CSAT scores, retrieval precision, and measurable task completion rates in production systems.
- Full-stack engineering capability in Python and TypeScript - building FastAPI/Express microservices, containerized deployments with Docker/Kubernetes, and production-grade CI/CD pipelines with GitHub Actions.
- Natural communicator who bridges AI research and engineering execution - translating complex model behavior into production-safe systems with business-aligned tradeoffs, clear stakeholder reporting, and delivery credibility.
- Proven business impact: $42K+ annual inference cost reduction · $1.8M operational savings · prior auth approval cycle reduced from 72 hours to 8 hours · entity extraction F1 improved from 73% to 89%.

TECHNICAL SKILLS

AI Agents & LLMs	Multi-agent architectures, Tool-calling workflows, LangChain, LangGraph, LlamaIndex, GPT-4, Claude, LLaMA 2, LoRA fine-tuning, RLHF, Prompt Engineering
RAG & Vector Search	Document ingestion pipelines, Chunking strategies, Retrieval optimization, Pinecone, FAISS, Retrieval evaluation frameworks
Backend & Languages	Python, TypeScript, Node.js, FastAPI, Express.js, RESTful APIs, Microservices
ML Frameworks	PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, Optuna, Dask
Cloud & Infrastructure	AWS (primary): SageMaker, Bedrock, ECS, Lambda, RDS, ElastiCache, CDK, CloudWatch Azure: AI Foundry, Databricks, OpenAI Service, AKS
MLOps & DevOps	Docker, Kubernetes, MLflow, Kubeflow, Weights & Biases, GitHub Actions, Terraform, CI/CD
Databases	PostgreSQL, Redis, MongoDB, DynamoDB
Observability & Quality	SLI/SLO-driven monitoring, Structured logging, A/B evaluation frameworks, Pytest, Jest, Load testing

EXPERIENCE

Senior AI Engineer **August 2025 – Present**
Northern Trust Bank – Dallas, TX

- Architected end-to-end conversational AI agent for investment advisory using GPT-4, LangChain, and RAG over 150K+ proprietary financial documents - serving 85K+ clients at 92% answer accuracy with sub-200ms p95 latency at 12K concurrent users.
- Designed a modular agent platform separating retrieval, reasoning, and execution layers for independent scaling and safer incremental rollout - enabling zero-downtime feature deployments and faster iteration cycles across 5 product squads.
- Implemented circuit-breaker patterns, retry logic, and fallback retrieval strategies across all agent components - maintaining 99.8% uptime under upstream LLM failures and spike traffic with no SLA breaches since launch.
- Built an intelligent LLM routing layer in TypeScript selecting between GPT-4 and GPT-3.5-turbo based on real-time query complexity scoring - saving $42K/year in inference costs while maintaining answer quality benchmarks.
- Fine-tuned LLaMA 2 (7B) using PyTorch and LoRA on 50K+ compliance documents with RLHF signals from 250+ domain analysts - lifting entity extraction F1 from 73% to 89%, reducing manual review time by 60%, and cutting fraud false positives by 38%.
- Deployed complete infrastructure-as-code via AWS CDK across ECS, Lambda, RDS, and ElastiCache with Redis response caching - reducing infrastructure provisioning time by 70% and enabling one-click environment replication.
- Established SLI/SLO observability framework with CloudWatch dashboards tracking latency p95 <200ms, error rate <0.5%, and availability >99.5% - enabled on-call engineers to diagnose and resolve incidents 45% faster.
- Enforced PCI-DSS governance across all AI workloads - implemented audit logging, bias detection on 100% of LLM outputs, data residency controls, and quarterly compliance reviews with zero regulatory findings.
- Achieved >90% test coverage across unit, integration, and load tests - introduced load testing suite simulating 15K concurrent users to validate latency SLOs ahead of every production release.
- Led architecture review and knowledge transfer sessions for a team of 6 engineers - documented design decisions, runbooks, and prompt engineering guidelines adopted as internal standards across the AI platform group.
- Benchmarked and selected vector store strategy (Pinecone vs. FAISS vs. pgvector) through systematic retrieval precision experiments - final choice improved recall@10 by 14% while reducing query latency by 22%.
- Collaborated with product, compliance, and quantitative research teams to define agent evaluation criteria - built automated regression suite catching prompt drift before production, reducing rollback events by 55%.

Environment: Python, TypeScript, GPT-4, GPT-3.5-turbo, LangChain, LangGraph, LLaMA 2, LoRA, RLHF, PyTorch, Hugging Face Transformers, RAG, Pinecone, FAISS, FastAPI, Node.js, AWS CDK, ECS, Lambda, RDS, ElastiCache, Redis, CloudWatch, Docker, Kubernetes, GitHub Actions, PCI-DSS

AI/ML Engineer **January 2024 - August 2025**
BlueCross Blue Shield(BCBS)– Kansas City, KS

- Built and deployed an automated prior authorization AI agent using GPT-3.5 and RAG over 50K+ clinical policy PDFs - reduced approval cycle from 72 hours to 8 hours, eliminating manual overhead and saving $1.8M annually in operational costs.
- Developed BERT-based medical claims classification system processing 800K+ claims monthly at 94% accuracy - using long-lived stateful agents with cross-session context retention, workflow checkpoints, and multi-step review handling.
- Shipped a rigorous A/B evaluation framework for prompt and agent behavior variations across 45K+ member interactions -improved chatbot CSAT score from 3.2 to 4.6/5 over 6 months through iterative prompt optimization.
- Benchmarked embedding strategies across multiple vector DB configurations - reduced vector database costs by 63% while maintaining 91% retrieval precision, validated through offline and online evaluation.
- Deployed secure multi-tenant document ingestion pipeline with Azure Content Safety API guardrails - blocked 99.7% of PHI leak attempts with zero HIPAA violations across 18 months of production operation.
- Orchestrated MLOps lifecycle with Kubeflow and GitHub Actions - automated model retraining, versioning, and deployment, reducing stale model incidents by 78% and cutting release cycle time from 2 weeks to 3 days.
- Designed guardrail and safety layers using Azure AI Foundry and Azure OpenAI Service - implemented content filtering, PII redaction, and safety classification across all member-facing interactions, reviewed quarterly with compliance teams.
- Built real-time monitoring dashboards on Azure Monitor and Application Insights - tracked model drift, accuracy degradation, and user feedback signals, enabling proactive retraining before production quality dropped below thresholds.

Environment: Python, GPT-3.5, BERT, Azure OpenAI Service, Azure AI Foundry, Azure Content Safety API, LangChain, RAG, Kubeflow, MLflow, GitHub Actions, AKS, Databricks, PostgreSQL, Pinecone, FastAPI, Docker, Azure Monitor, HIPAA compliance

ML Engineer **January 2023 – December 2023**
Molina Health – Kansas City, KS

- Built DistilBERT + FAISS question-answering system handling 180K+ member eligibility queries monthly - achieved 88% accuracy and 180ms p95 latency, owning the full stack from document indexing through inference API and production monitoring.
- Engineered a care recommendation engine using collaborative filtering and gradient boosting - personalized plans for 450K+ Medicaid enrollees, improving member engagement by 29% and increasing plan retention metrics.
- Reduced model training time from 6 hours to 45 minutes using Dask distributed computing - enabling faster experimentation cycles and more frequent model refreshes without additional infrastructure cost.
- Built Apache Airflow ETL pipelines processing 2.3M patient records daily from 7 state Medicaid exchanges - maintained 99.6% data quality with automated validation, deduplication, and anomaly detection.
- Deployed containerized ML services using Terraform on AWS - configured auto-scaling groups achieving 99.9% uptime, handling peak eligibility query volumes during open enrollment periods without manual intervention.
- Maintained HIPAA compliance across all data pipelines - implemented field-level encryption, audit trails, role-based access controls, and quarterly data governance reviews with zero compliance findings.
- Implemented model performance monitoring with automated alerting - tracked prediction drift, data distribution shifts, and accuracy degradation, triggering retraining pipelines proactively before SLA violations occurred.
- Collaborated with clinical and operations teams to validate recommendation quality - ran user acceptance testing with care managers across 3 state offices, incorporating feedback to improve recommendation relevance by 21%.

Environment: Python, DistilBERT, FAISS, PyTorch, Scikit-learn, Gradient Boosting, Collaborative Filtering, Dask, Apache Airflow, AWS SageMaker, EC2, S3, Terraform, Docker, PostgreSQL, MLflow, GitHub Actions, HIPAA compliance

ML Engineer **May 2018 - July 2022**
NxtWave - Early-Stage, Seed Funded Startup - Hyderabad, TS

- Architected a content recommendation system using matrix factorization and neural collaborative filtering - increased course completion rates by 26% for 120K+ learners, designed, deployed, and iterated independently with full ownership.
- Built a transformer-based auto-grading agent evaluating 45K+ code submissions weekly at 91% human-parity accuracy - reducing grading turnaround from 24 hours to under 2 hours at 3x submission volume growth.
- Developed an adaptive learning path engine using contextual bandits - reduced learner dropout rates by 18% by dynamically adjusting content sequencing based on real-time engagement and performance signals.
- Improved recommendation model AUC from 0.78 to 0.87 using Optuna hyperparameter optimization - ran systematic experiments across model architectures, feature sets, and training strategies with reproducible MLflow tracking.
- Designed and maintained data ingestion and feature engineering pipelines for learner behavior data - processing clickstream, assessment, and video engagement signals at scale to feed downstream ML models.
- Collaborated with curriculum designers and product teams to define ML evaluation metrics aligned with learning outcomes - established offline and online evaluation protocols adopted as product team standards.

Environment: Python, PyTorch, TensorFlow, Scikit-learn, Matrix Factorization, Neural Collaborative Filtering, Contextual Bandits, Optuna, MLflow, Hugging Face Transformers, FastAPI, PostgreSQL, Redis, Docker, AWS EC2, S3, GitHub Actions

EDUCATION

Master's in computer science, University of Central Missouri, Missouri, US **August 2022 - May 2024**
Bachelor's in computer science, CMS College of Engineering, TN, India **June 2015 - May 2019**

CERTIFICATION

- AWS Certified Machine Learning - Specialty | 2024
- Google Cloud Professional Machine Learning Engineer | 2024
- Microsoft Certified: Azure AI Engineer Associate | 2023
- DeepLearning.AI - LLM Fine-Tuning Specialization | 2024